

Explaining National Volleyball Success Through Economic and Development Indicators

1. Motivation

I have been following volleyball for years. Watching Nations League matches and World Championships, I kept noticing something that didn't quite add up: some of the wealthiest countries in the world were nowhere near the podium, while Brazil and Turkey were almost always in the final rounds despite having much smaller economies. At the same time, countries with enormous populations like India or Nigeria weren't dominating either, which made me question whether raw numbers like GDP or population actually tell us anything meaningful about athletic success.

Turkey is a particularly interesting case. I grew up watching Vakıfbank and Fenerbahçe dominate European club volleyball, and the national women's team consistently finishing in the top four at World Championships, ahead of countries with GDP per capita two or three times higher. That gap between economic indicators and volleyball results was what originally made me want to look at this properly.

The central question I wanted to answer was whether national development and economic indicators (GDP per capita, HDI, population) are associated with volleyball success and if so, whether the relationship looks different for men's and women's teams.

2. Data

Volleyball ranking data (Senior Men's and Women's World Rankings, 2025) was collected from Wikipedia, which mirrors the official FIVB data. The FIVB website renders its ranking tables via JavaScript and doesn't support direct export, so Wikipedia was the most reliable source for exact point values.

Economic data came from three publicly available sources: GDP per capita and total population from the World Bank (2023 values), and the Human Development Index from the UNDP Human Development Report (2023). All four sources were merged on standardised country names, resulting in a final dataset of 38 countries.

GDP per capita and population were log-transformed before any analysis. Both are heavily right-skewed – a handful of very wealthy or very populous countries would otherwise distort regression results – and after the transformation the distributions become much more symmetric.

3. Analysis and Findings

Before focusing on individual findings, I first checked the correlation matrix to get a general idea of the dataset. I wanted to see whether GDP, HDI, or population seemed more related to volleyball ranking points. This helped me decide which patterns were worth discussing in more detail. As Figure 1 shows, HDI seems to have the clearest positive relationship with volleyball success, while population does not show a strong connection.

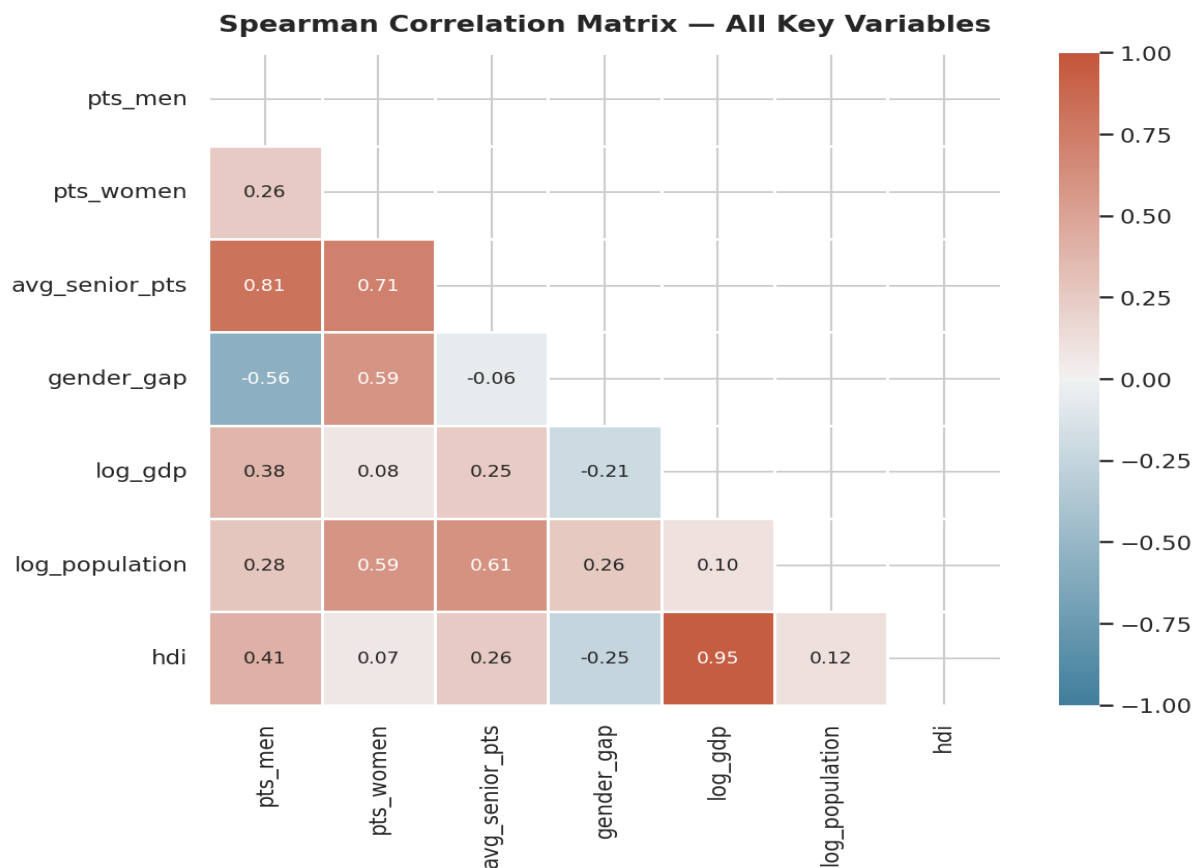


Figure 1. Spearman correlation matrix between socioeconomic indicators and volleyball performance variables. HDI has the clearest positive relationship with volleyball performance, while population has a much weaker relationship

The HDI-Volleyball Relationship

The clearest finding across the entire project is that HDI is a stronger predictor of volleyball success than GDP. The Spearman correlation between HDI and women's WR points is around 0.50 ($p < 0.05$), compared to roughly 0.40 for log GDP. This pattern holds consistently across correlation analysis, regression coefficients, and Decision Tree feature importance the fact that HDI always comes out on top.

What makes this interesting is that HDI captures something broader than just income: it combines health, education, and living standards into a single number. The finding suggests that volleyball success is tied more to overall societal development than to raw economic output. A country can be moderately wealthy but still produce elite volleyball if its broader development infrastructure is strong.

The HDI effect is also noticeably larger for women's volleyball than for men's. In the multiple regression models, the HDI coefficient is higher on the women's side, which suggests women's volleyball performance is particularly sensitive to national development levels — possibly because women's sports programs tend to require stronger institutional support and gender-inclusive development to thrive.

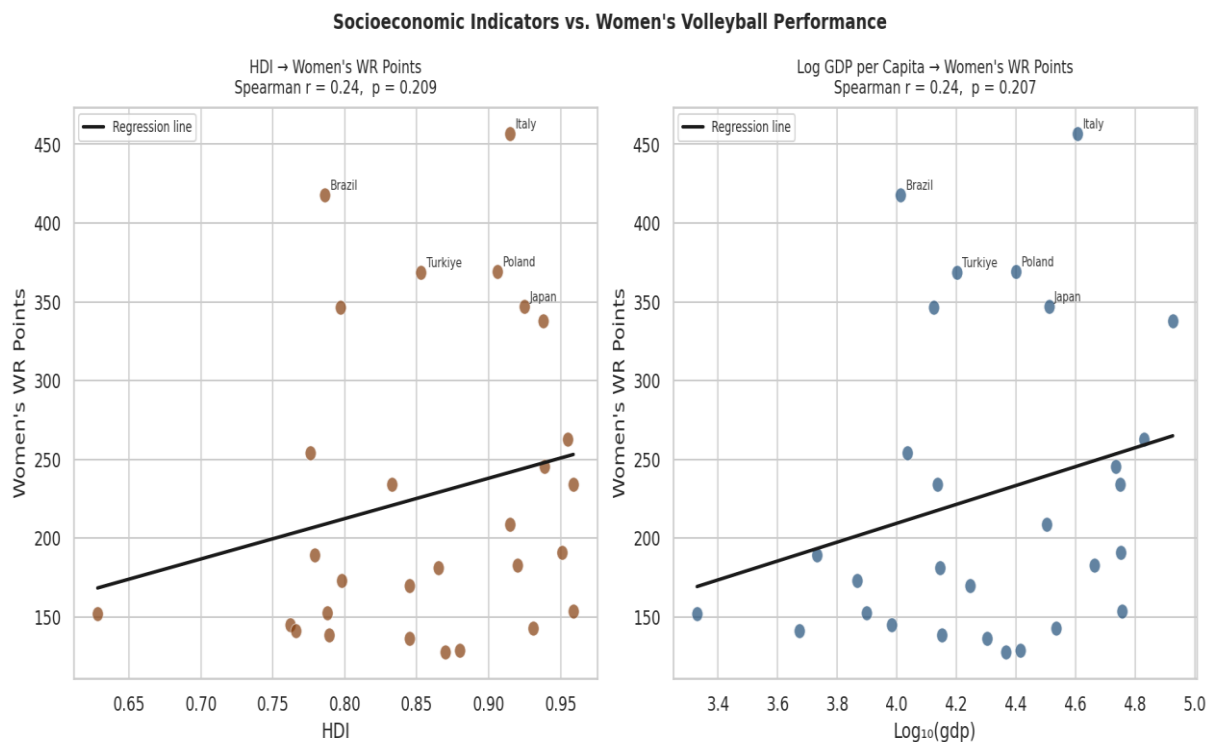


Figure 2. Relationship between HDI and women's volleyball ranking points. Countries with higher HDI generally tend to have stronger women's volleyball performance.

The Population Paradox

There is essentially no statistical relationship between population size and volleyball success (Spearman $r \approx 0.10$, not significant). Slovenia, with around 2 million people, consistently ranks in the global top five. Poland, with 38 million, is currently the world number one men's team. Indonesia has 270 million people and doesn't make the top 30.

This was actually one of the more surprising results. The intuition that more people means a bigger talent pool to draw from turns out not to hold, at least not at the country level. What

seems to matter much more is the quality of the development pipeline, not the raw number of potential athletes.

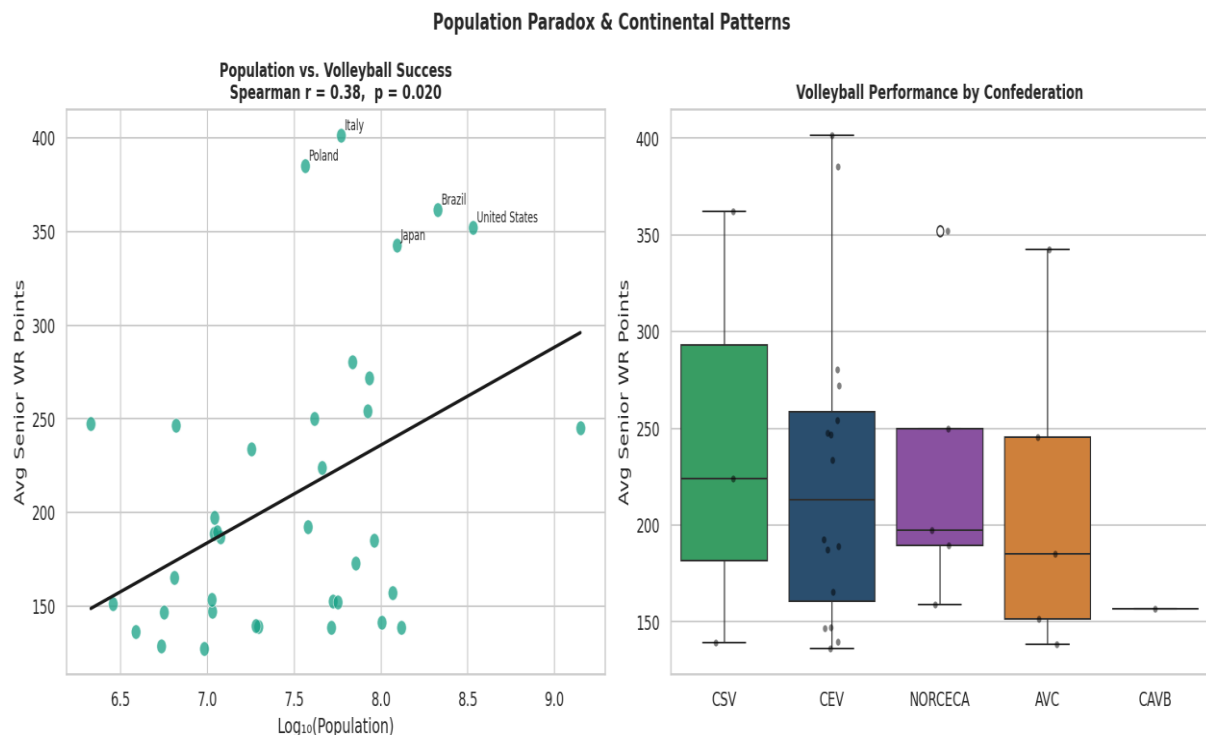


Figure 3. Population size shows almost no clear relationship with volleyball success. Smaller countries such as Slovenia can outperform far more populous nations, suggesting that development structure and sports culture matter more than raw population size.

Over-Performers: Brazil and Turkey

The residual analysis was probably the most insightful part of the project. After fitting a regression model using log GDP to predict women's WR points, the residuals, the gap between what the model predicted and what the actual ranking was consistently flagged Brazil and Turkey as the biggest positive outliers. Their results far exceed what their economic indicators would suggest.

Brazil's GDP per capita is around 10,000 USD, roughly comparable to countries that don't make the top 30. Turkey sits at about 13,000 USD. Yet both have women's teams consistently ranked in the global top five.

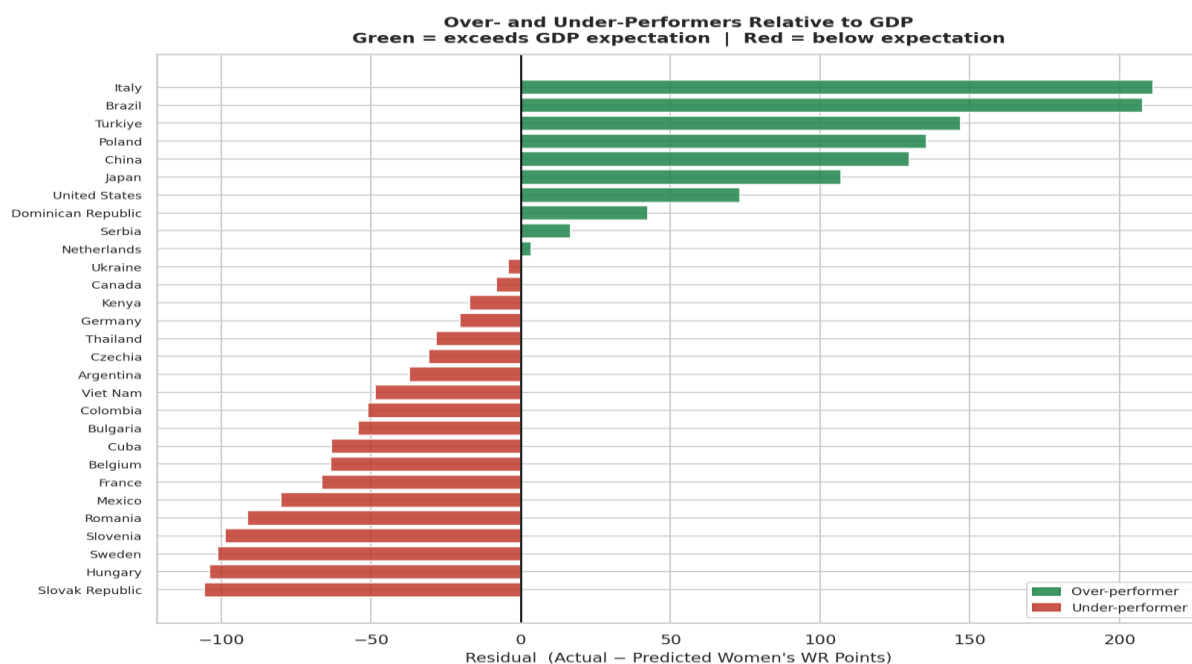


Figure 4. Residual analysis of women's volleyball ranking points based on GDP. Brazil and Turkey appear as strong positive outliers.

The data can't explain this fully, which is itself a meaningful finding: something beyond GDP and HDI is at work club infrastructure, federation investment, the historical tradition of volleyball in both countries, and the fact that the sport carries genuine cultural weight there in a way it doesn't everywhere.

On the Regression Models

The regression models predicting exact WR points showed low R^2 values, and in some test splits the scores dropped below zero which the meaning the model performed worse than simply predicting the mean. This was initially disappointing but makes sense on the reflection. With 38 countries and high variance in volleyball performance, GDP, HDI and population can set a broad context but can't reliably predict exact ranking points. A large portion of what makes a country good at volleyball simply isn't captured in macroeconomic data.

The clustering results were more informative here. K-Means and hierarchical clustering both identified the same natural groupings: a cluster of wealthy, high-HDI, strong-volleyball European nations; a cluster of developing countries with emerging programs; and a cluster containing Brazil and Turkey, which sit apart from the others precisely because their volleyball performance is disproportionate to their economic profile.

Confederation Classification

Since regression on exact points didn't work well, the prediction task was reformulated: instead of predicting WR points, can we predict which confederation a country belongs to based on its socioeconomic profile? This turned out to be a much better fit. Logistic Regression, kNN, and Decision Tree classifiers all achieved accuracy well above the 25% random baseline (around 56% for Logistic Regression with 5-fold cross-validation). CEV (European) countries were the easiest to classify, since they have a distinct combination of high GDP, high HDI, and strong volleyball performance. NORCECA and AVC countries overlap more in economic space, which made them harder to separate.

4. Overall Answer to the Research Question

Based on the results, I would say that GDP, HDI and population explain some parts of national volleyball success, but they are not enough to explain the full picture. Figure 1 gives the general overview of the relationships in the dataset, and the strongest pattern appears around HDI rather than population. This supports the idea that volleyball success is more related to broader development conditions than just having a large economy or a large population.

Among the three main indicators, HDI seems to be the most meaningful one, especially for women's volleyball. Figure 2 also supports this because countries with higher HDI generally tend to have higher women's ranking points. GDP has some relationship too, but it is not as clear as HDI. Population, on the other hand, appears to be the weakest variable. As shown in Figure 3, countries with smaller populations can still perform very well, while some very populous countries do not appear among the strongest volleyball nations. This means that a bigger talent pool does not automatically create a stronger national team.

The Brazil and Turkey result also shows why socioeconomic indicators alone are not enough. In Figure 4, both countries appear as positive outliers, meaning that their women's volleyball performance is higher than what the GDP-based model would expect. This suggests that other factors such as club systems, volleyball culture, federation support, and the popularity of the sport probably play an important role.

So, my final answer is not a simple yes or no. Socioeconomic indicators do matter, especially HDI, but they only explain part of national volleyball success. The rest seems to depend on sport-specific and cultural factors that are not directly captured by GDP, HDI, or population data.

5. Limitations and Future Work

The main limitation is sample size. 38 countries is small for machine learning, and the learning curve analysis in the regression notebooks shows the models haven't fully converged and more data would likely improve performance. A time-series dataset going back to 2010 or earlier would be particularly valuable: it would allow tracking whether changes in HDI or GDP over time actually predict changes in volleyball rankings, rather than just looking at a single snapshot.

The biggest unmeasured variable is volleyball culture and federation quality which are the things that clearly matter but don't appear in any publicly available dataset. Brazil and Turkey being the biggest outliers in the residual analysis is a good illustration of this gap.

Future extensions could include adding sports-specific predictors (national sports budgets, number of professional leagues, Olympic medal history), applying Ridge or Lasso regression to handle the correlation between GDP and HDI, and extending the analysis to youth categories once reliable FIVB ranking data becomes available.

5. AI Usage Disclosure

AI assistance (Claude, Anthropic) was used in certain parts of this project alongside course materials. The analysis methods were primarily based on course recitations and lecture notes. AI was used mainly for debugging errors during data loading (zip file parsing issues, HDI column detection, IndexError in World Bank data), structuring the notebook layout, README formatting, and minor visualisation adjustments. The research questions, data collection decisions, method choices, interpretation of results, and all conclusions are my own work.